

CS410 Final

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I. QUESTION (MIN: 4 MAX: 20)

Let $\Sigma = \{a, b\}$. Let C_4 be the language consisting of all strings that contain an 'a' exactly 4 places from the right end. Draw a DFA that recognizes C_4 .

II. QUESTION (MIN 3 MAX: 15)

Let $A_5 = \{a^p b^q c^r \mid \text{where } p + q + r \text{ is a multiple of } 5\}$. Draw an NFA that recognizes A_5 .

III. QUESTION (MIN: 4 MAX: 20)

Give the CFG for the following language: $D = \{w\#x \mid w^R \text{ is a substring of } x \text{ for } w, x \in \{0, 1\}^*\}$

IV. QUESTION (MIN: 3 MAX: 15)

Give a PDA generating the language of strings with three times as many a's as b's. The alphabet is $\{a, b\}$.

V. QUESTION (MIN: 6 MAX: 30)

Design a single tape TM that recognizes the language $L = \{0^n 1^{n^2} 2^{n^2} \mid n \geq 1\}$. Draw a state diagram, give states, transitions, etc.